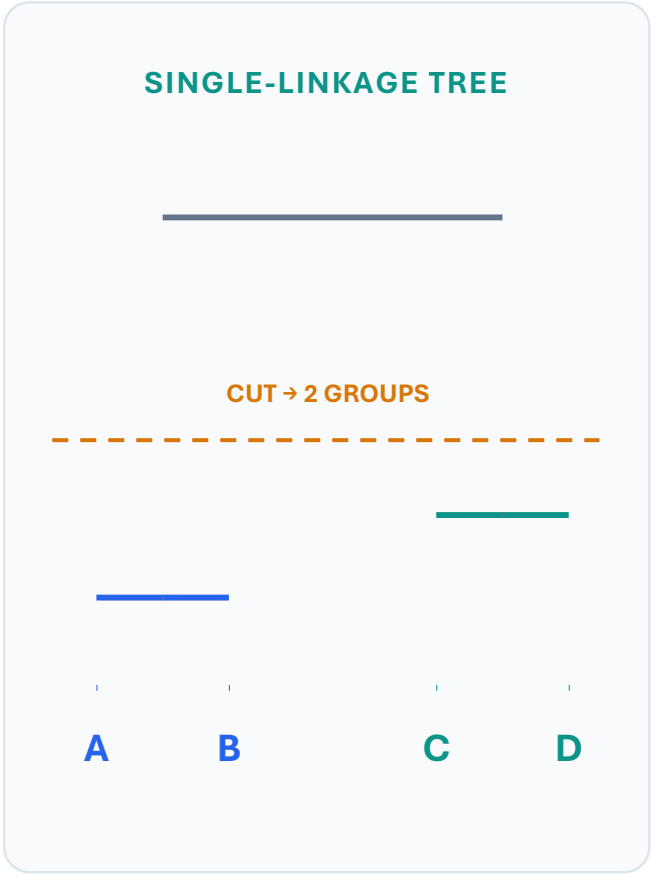
UNSUPERVISED LEARNING

Hierarchical Clustering

From distances and merges to dendrograms and choosing a cut

WORKED EXAMPLES

PYTHON + SCIKIT-LEARN



LESSON MAP

What will we learn?

One continuous example, then a real dataset.

Start with points 01

Each point is its own cluster.

Calculate distances 02

Build a distance matrix.

Merge using linkage 03

Combine the closest clusters.

Read dendrogram 04

See which groups merged and when.

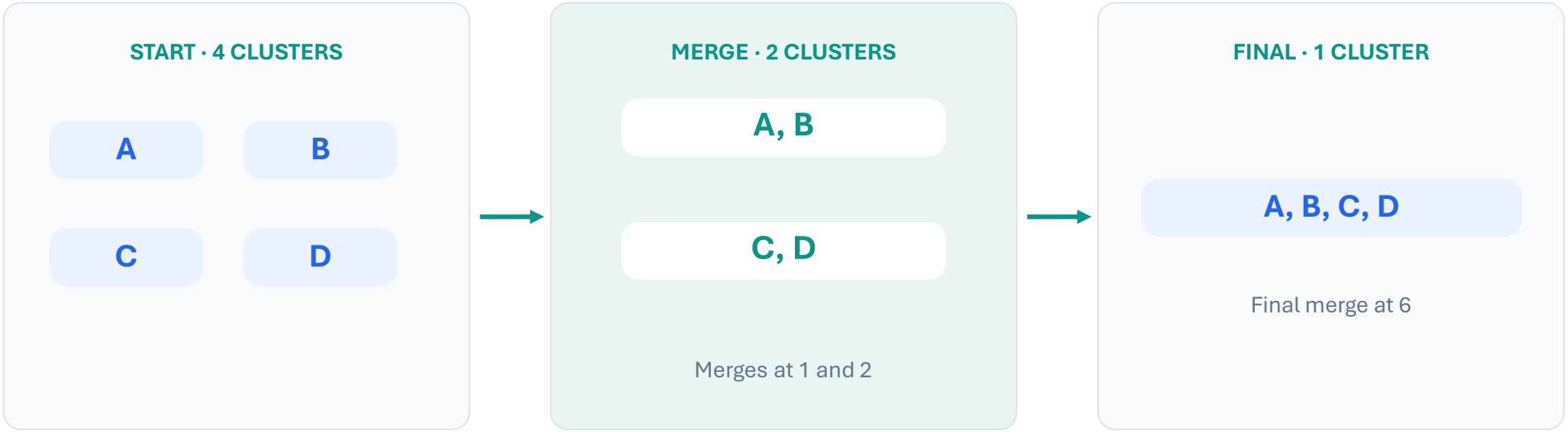
Choose K and apply 05

Select K; implement with Python.

GOAL **Interpret a dendrogram and select meaningful groups.**

Hierarchical clustering builds a tree of groups

Agglomerative clustering repeatedly merges the nearest clusters.

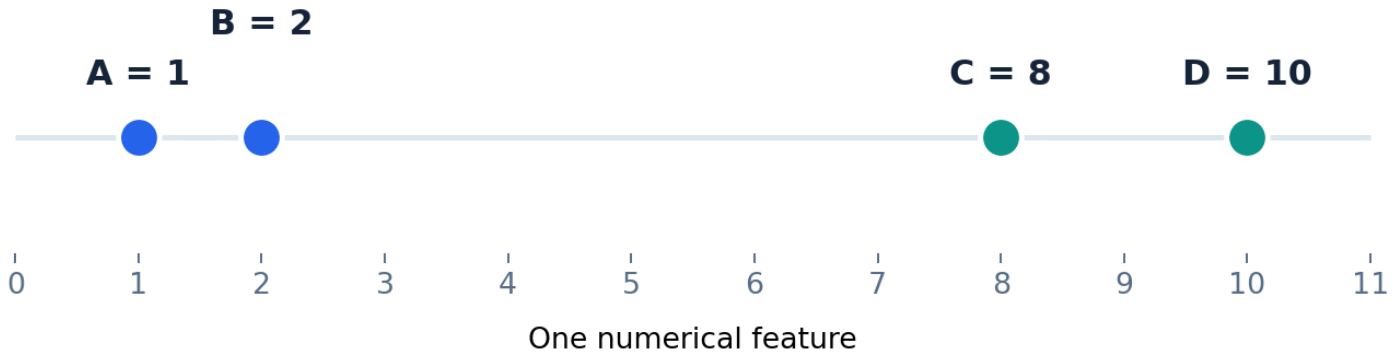


REMEMBER **Bottom-up: begin with individual points, then merge clusters.**

Use the same four points throughout

One numerical feature keeps our first calculations simple.

Point	Value
A	1
B	2
C	8
D	10



FIRST **Each point starts as its own cluster: A | B | C | D.**

Calculate the distance between each pair

For one feature, distance = absolute difference between values.

	A (1)	B (2)	C (8)	D (10)
A (1)	0	1	7	9
B (2)	1	0	6	8
C (8)	7	6	0	2
D (10)	9	8	2	0

A to B

$|1 - 2| = 1$

C to D

$|8 - 10| = 2$

OBSERVE **The closest pair is A and B (distance 1).**

Merge A and B at distance 1

Always merge the closest two current clusters.



Calculation

Distance(A, B) = |1 - 2| = 1

Clusters remaining

(A, B) | C | D

STATUS **4 clusters become 3 clusters.**

Next, merge C and D at distance 2

Compare distances between the three remaining clusters.

Remaining cluster pair	Single-linkage distance
(A, B) to C	min(7, 6) = 6
(A, B) to D	min(9, 8) = 8
C to D	|8 - 10| = 2

Cluster 1: A, B

Cluster 2: C, D

STATUS **3 clusters become 2 clusters.**

How far apart are Cluster 1 and Cluster 2?

Cluster 1 = A, B | Cluster 2 = C, D

Cross-cluster pair	Calculation	Distance
A to C	|1 - 8|	7
A to D	|1 - 10|	9
B to C	|2 - 8|	6
B to D	|2 - 10|	8

Single Linkage

Choose the minimum distance.

min(7, 9, 6, 8) = 6

RESULT **The two clusters merge at distance 6 using Single Linkage.**

Different methods define cluster distance differently

Use the same four cross-cluster distances: 7, 9, 6, 8.

Single Linkage

Nearest pair

min(7, 9, 6, 8)

Distance = 6

Complete Linkage

Farthest pair

max(7, 9, 6, 8)

Distance = 9

Average Linkage

Mean of all pairs

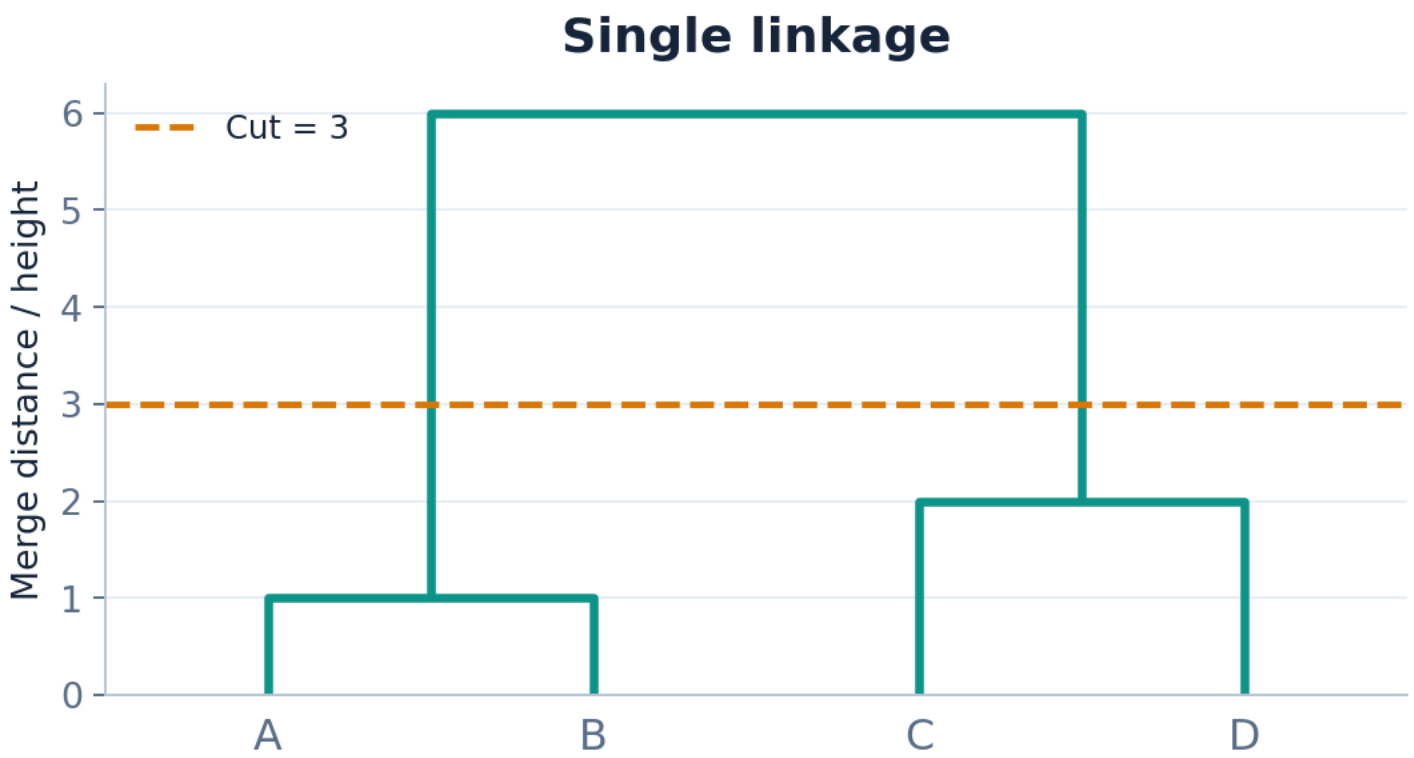
(7 + 9 + 6 + 8) / 4

Distance = 7.5

WHY IT MATTERS **Linkage can change merge heights and sometimes cluster membership.**

A dendrogram records every merge

Merge height = distance reported by the chosen linkage method.



Read from the bottom

A + B: height 1
C + D: height 2
Two groups: height 6

INTERPRET **A cut strictly between heights 2 and 6 gives two clusters.**

Lower cut → more clusters; higher cut → fewer

A horizontal line decides which merges are included.



RULE **Count the branches crossed by the horizontal cut line.**

Look for a large jump in merging distances

For the four-point data: 1 → 2 → 6.

Merge	Height	Jump from previous
A with B	1	—
C with D	2	1
(A, B) with (C, D)	6	4

Choose a cut

Cut at height 3

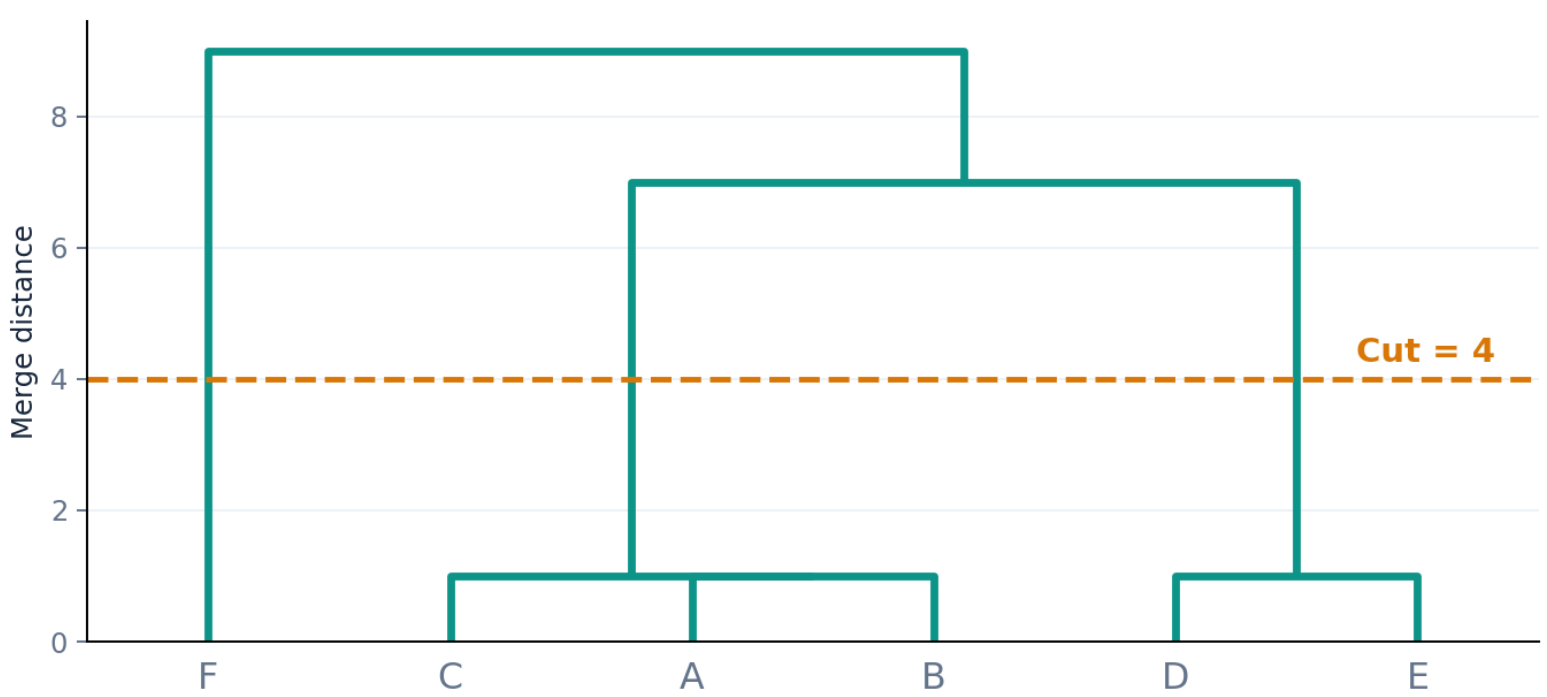
Result: 2 clusters

Largest gap is between heights 2 and 6.

CAUTION **A large jump is a useful clue, not a guarantee.**

Six points: where should we cut?

Values: A=1, B=2, C=3, D=10, E=11, F=20.



Observe the gap

Merge heights:
1, 1, 1, 7, 9

Cut at 4
→ 3 clusters

GROUPS **Cluster 1: A, B, C | Cluster 2: D, E | Cluster 3: F**

What if we already know the number of clusters?

Choose a horizontal cut that crosses exactly K branches.

Business requirement: K = 3

Cut height for six-point data	Clusters
4	A,B,C | D,E | F
8	A,B,C,D,E | F
10	A,B,C,D,E,F

DECISION **For K = 3, choose any cut strictly between heights 1 and 7.**